

STAT 24620=FINM 34700, STAT 32950
Multivariate Data Analysis
Lecture 14: Nonlinear Unsupervised Learning

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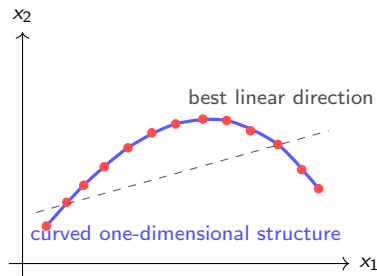
Outline

1 Principal curves

2 Distance-Based Embeddings

When linear PCA is not enough

- PCA summarizes data using a low-dimensional **linear structure**.
- This works well when variation is approximately elliptical or planar.
- But some datasets have curved geometry:
 - images of a face under different rotations, where the images vary smoothly but not along a linear subspace of the pixel space;
 - points lying near a curved trajectory or spiral in a high-dimensional space.
- A linear projection may fail to follow the intrinsic curved structure of the data.



Can we generalize PCA by replacing a linear subspace with a smooth curve?

Principal curves: a nonlinear first component

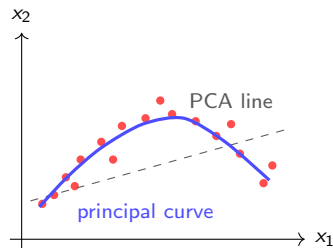
- PCA summarizes data by a straight line:

$$x_i \approx \mu + z_i v_1.$$

- A principal curve replaces the line by a smooth curve:

$$x_i \approx f(t_i), \quad t_i \in \mathbb{R}.$$

- Think of t_i as a one-dimensional coordinate along the curve.
- Useful when the main variation follows a trajectory rather than a line.



Principal curves are a direct nonlinear analogue of the first principal component.

Fitting a principal curve

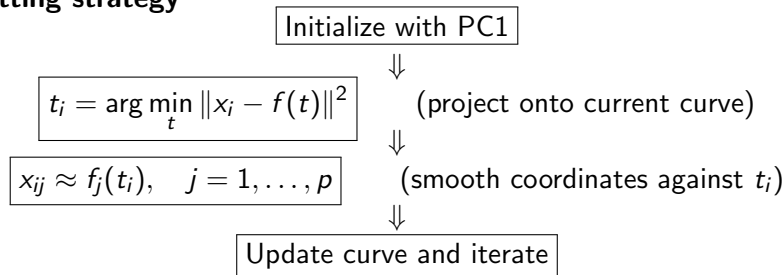
Let $f(t) \in \mathbb{R}^p$ be a smooth curve, and let t_i denote the location of x_i along the curve.

We would like to solve

$$\min_{f, t_1, \dots, t_n} \sum_{i=1}^n \|x_i - f(t_i)\|^2 + \lambda \text{roughness}(f).$$

- Fit the data well
- Keep the curve smooth

Alternating fitting strategy



Another perspective: geometry

Principal curves

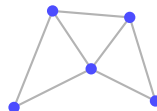
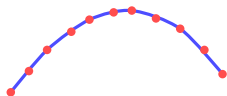
$$x_i \approx f(t_i)$$



Geometry-based methods

preserve distances or neighborhoods

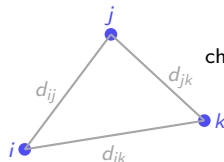
learn a smooth nonlinear structure
directly



What does it mean for two points to be close?

MDS: preserving pairwise distances

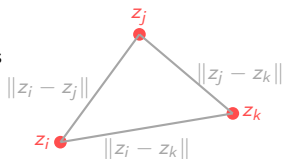
Distance information



choose coordinates



Low-dimensional map

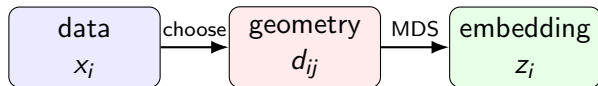


$$\|z_i - z_j\| \approx d_{ij}$$

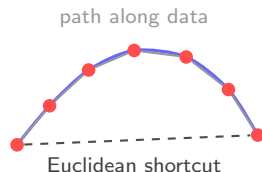
- Classical MDS finds coordinates that preserve pairwise distances.
- With Euclidean distances, classical MDS gives essentially the same representation as PCA (up to rotation/reflection).
- Intuitively, PCA also preserves large-scale Euclidean geometry: distant points tend to remain distant after projection.
- Nonlinear methods often keep the embedding step but replace Euclidean distance by a better notion of distance.

Changing the distance

- MDS separates the problem into two steps: choose distances, then embed.



- For curved structure, Euclidean distance can create a shortcut through empty space.
- A nonlinear route: replace Euclidean distance by distance along the data.



This motivates geodesic distance.

The key question: what distance captures the geometry of the data?

Isomap = MDS with geodesic distance

Isomap keeps the MDS embedding step, but changes the distance matrix.

- 1 Build a neighbor graph:

$$i \sim j \quad \text{if } x_j \text{ is among the } k \text{ nearest neighbors of } x_i.$$

- 2 Put edge weights equal to Euclidean distances between neighbors.
- 3 Approximate geodesic distances by shortest-path distances on the graph:

$$d_{ij}^{\text{geo}} = \text{shortest path distance from } i \text{ to } j.$$

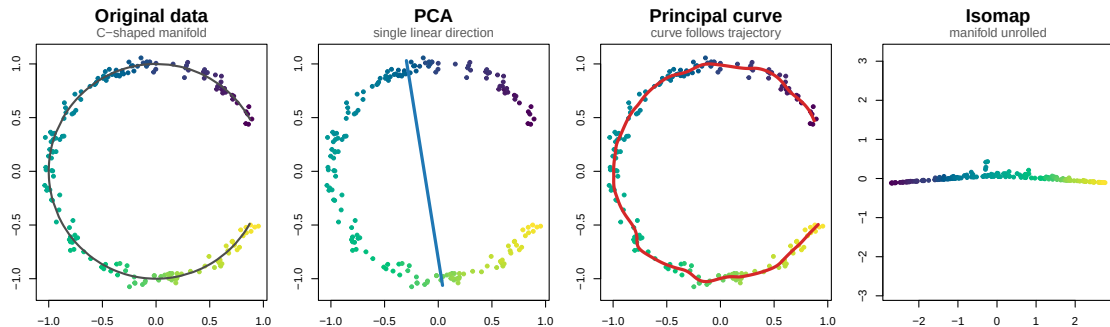
- 4 Apply classical MDS to D^{geo} .

Isomap = MDS using graph-based geodesic distances.

When does Isomap work well?

- Isomap works well when the data lie near a smooth low-dimensional manifold.
- The neighborhood graph should capture local geometry without creating shortcuts.
- Choosing the neighborhood size k is important:
 - too small: graph disconnects;
 - too large: curvature is lost.
- Isomap is less effective for disconnected clusters or highly noisy data.

Simulation comparison



- PCA gives the best linear direction, but it cannot follow the curved trajectory.
- A principal curve estimates the nonlinear trajectory directly.
- Isomap uses graph distances and tries to unroll the manifold in the embedding.

Summary

- PCA is linear; it can miss curved low-dimensional structure.
- Principal curves replace the first PC line by a smooth one-dimensional curve.
- MDS makes the role of pairwise distances explicit.
- Classical MDS with Euclidean distances gives a PCA-like embedding.
- Isomap replaces Euclidean distances with graph-based geodesic distances.
- The main tuning choice in Isomap is the neighborhood graph.

Changing the distance changes the geometry that the embedding tries to preserve.